USN | | | | | | | | | | |

[23CSE304]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

DATABASE MANAGEMENT SYSTEMS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			5 x2=10
1.	a.	What is a weak and strong entity set? Give Example.	BTL 2	CO1	02
	b.	Write a relational algebra expression to select all employee tuples from the EMPLOYEE relation where the department number is 10.	BTL 3	CO2	02
	c.	list the aggregate functions supported by SQL	BTL 1	C03	02
	d.	What is a trigger in SQL.?	BTL 1	CO4	02
	e.	Define BCNF.	BTL 2	CO5	02
		PART-B			**6 X 5=30**
2.	a.	What is an Attribute? Explain different types of Attributes.	BTL 2	CO1	05
		OR			
	b.	List the advantages of database approach.	BTL 1	CO1	05
3.	a.	What is a Primary Key? Explain some desirable Primary Key characteristics.	BTL 2	CO2	05
		OR			
	b.	Discuss Entity Integrity constraint with an example.	BTL 3	CO2	05
4.	a.	Write an SQL query to i. Find the maximum, minimum, and average salary of the employees. ii. To find the employee id whose salary lies in the range of 50000 and 150000.	BTL 3	CO3	05
		OR			
	b.	Illustrate Group by and having clauses in SQL with examples	BTL 3	CO3	05
5.	a.	Explain Nested Queries and correlated queries.	BTL 2	CO4	05
		OR			
	b.	What is a Join? Discuss Equi Join and Natural joins with example.	BTL 2	CO4	05
6.	a.	Explain different Anomalies in Designing a Database	BTL 2	CO5	05
		OR			
	b.	Explain different types of Functional dependency.	BTL 2	CO5	05
7.	a.	Write notes on Wild card operators in SQL.	BTL 3	CO3	05
		OR			
	b.	Discuss the following terms: i. Entity , Attribute ii. Candidate key , Foreign key	BTL 2	CO2	05

JGU (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagar-112

		PART-C			**6 X 10=60**
8.	**a.**	Explain the Three schema architecture of DBMS with a neat sketch.	BTL 2	CO1	**10**
		OR			
	b.	Construct an E-R diagram for a Hospital Management system database with suitable entities and attributes.	BTL 3	CO1	**10**
9.	**a.**	Explain Relational Model concepts with example.	BTL 2	CO2	**10**
		OR			
	b.	List and describe the basic operations of relational algebra	BTL 2	CO2	**10**
10.	**a.**	List and explain various DML, DDL commands in SQL	BTL 2	CO3	**10**
		OR			
	b.	Define Alter, Drop and Rename commands in SQL with example	BTL 2	CO3	**10**
11.	**a.**	Explain the concept of Cursors using appropriate example	BTL 3	CO4	**10**
		OR			
	b.	What are views? How they are created? List the uses of View.	BTL 3	CO4	**10**
12.	**a.**	Explain the concept of Normalization and discuss its benefits	BTL 4	CO5	**10**
		OR			
	b.	Explain the concept of Concurrency control with examples	BTL 3	CO5	**10**
13.	**a.**	Consider a Table/relation called 'Student' with the following structure Column Names: Type Name: Char (20) USN: Varchar (10) Age: Integer Department: Char (10) Marks: Integer Write SQL queries to **i.** Retrieve the maximum marks of the students. **ii.** Retrieve the name and USN of the students those who belongs to CSE department with marks more than 70. **iii.** Retrieve all the student's details sorted by 'Age' in descending order. **iv.** Retrieve student details sorted by Marks in ascending order. **v.** Retrieve student details whose name start with 'S'.	BTL 5	CO3	**10**
		OR			
	b.	Discuss the ACID properties of the transaction management.	BTL 2	CO5	**10**